

Skills Summary

Limits at the Edges

Use this reference while completing your worksheet or when studying.

Skill 1 — Reading a Limit From One Side

Rule: $x \rightarrow k^-$ means $x < k$ and $x \rightarrow k^+$ means $x > k$. On one side, an absolute value or a piecewise rule becomes a single ordinary formula — rewrite it that way *first*, then take the limit. If the two one-sided values disagree, the two-sided limit does not exist.

Example: Evaluate $\lim_{x \rightarrow 1^-} \frac{|x-1|}{x-1}$

Step 1. Approaching from the left means $x < 1$, so $x - 1$ is negative and the absolute value must be rewritten as $-(x - 1)$ before anything else.

Step 2. For $x < 1$ the expression is $\frac{-(x-1)}{x-1} = -1$, a constant, so the limit is that constant.

$$\Rightarrow \lim_{x \rightarrow 1^-} \frac{|x-1|}{x-1} = -1$$

Try it: Evaluate $\lim_{x \rightarrow -2^+} \frac{|x+2|}{x+2}$

check: 1

Skill 2 — Hole or Vertical Asymptote?

Rule: at a zero of the denominator, factor and cancel. If the factor cancels, the point is a **hole** and the limit is a finite number — substitute into the reduced form. If it does not cancel, the point is a **vertical asymptote**: the limit is $+\infty$ or $-\infty$, and the sign comes from the sign of the surviving factor on the side you approach from.

Example: For $F(x) = \frac{x^2 - 9}{x^2 - 2x - 3}$ find $\lim_{x \rightarrow 3} F(x)$ and $\lim_{x \rightarrow -1^+} F(x)$.

Step 1. Both 3 and -1 zero the denominator, so factor first — only the factor that cancels gives a hole.

Step 2. $F(x) = \frac{(x-3)(x+3)}{(x-3)(x+1)} = \frac{x+3}{x+1}$. At $x = 3$ that is $\frac{6}{4}$; at $x \rightarrow -1^+$ the top tends to 2 and $x + 1$ is a small positive.

$$\Rightarrow \text{hole: } \frac{3}{2} \quad \text{asymptote: } +\infty$$

Try it: For $G(x) = \frac{x^2 - 1}{x^2 - 3x + 2}$ find $\lim_{x \rightarrow 1} G(x)$ and $\lim_{x \rightarrow 2^+} G(x)$.

check: $\infty+$ and $2-$

Skill 3 — End Behavior as x Grows Large

Rule: divide every term by the highest power of x in the *denominator*, then use $\frac{c}{x^n} \rightarrow 0$.
Shortcut for polynomials: top degree < bottom degree gives 0; equal degrees give the ratio of leading coefficients; top degree > bottom degree gives $\pm\infty$.

Example: Evaluate $\lim_{x \rightarrow \infty} \frac{4x^3 - x}{2x^3 + 5}$

Step 1. Top and bottom are both cubic, so divide through by x^3 and let every leftover fraction die.

Step 2. $\frac{4 - 1/x^2}{2 + 5/x^3} \rightarrow \frac{4 - 0}{2 + 0} = 2$, which is just the ratio of the leading coefficients.

$$\Rightarrow \lim_{x \rightarrow \infty} \frac{4x^3 - x}{2x^3 + 5} = \mathbf{2}$$

Try it: Evaluate $\lim_{x \rightarrow \infty} \frac{7x^3 + 2x}{2 - x^3}$

✓ — check

(!) Watch out: $\sqrt{x^2} = |x|$, not x . Heading to $-\infty$ that minus sign is real, so an expression like $\sqrt{9x^2 + 4} / (3x - 1)$ tends to 1 at the right end and -1 at the left end.
